MARYE AGEGN

Biomedical Engineer | Scalable Medical Devices & Digital Health | Healthcare Technology Specialist

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PROFESSIONAL & ACADEMIC SUMMARY

Biomedical engineer with over three years of frontline healthcare technology management and clinical engineering experience across public hospitals and healthcare systems in Ethiopia, advancing into graduate engineering research at Anna University. Specialized in building scalable medical devices, pre-procurement technical specification formulation, equipment commissioning, and preventive maintenance. Academic research focus encompasses biosignal processing, gait analysis, rehabilitation engineering, neuroimaging, and explainable AI in healthcare.

EDUCATION

Master of Engineering (M.Eng.) in Biomedical Engineering — *July 2025 – Present*

Anna University, Chennai, India

- Graduate research focus: Lower-back inertial sensing, mobility assessment, gait analysis, and biosignal processing.
- Coursework: Advanced Biosignal Analysis, Medical Device Standards, Healthcare AI, and Rehabilitation Systems.

Bachelor of Science (B.Sc.) in Biomedical Engineering — *2016 – 2021*

University of Gondar, Ethiopia — Cumulative GPA: 3.78 / 4.00 (Distinction)

- Capstone Thesis: Design of a Simple Low-Cost Repetitive Transcranial Magnetic Stimulation (rTMS) System for Major Depressive Disorder in Low-Resource Settings (Evaluated Grade: A).
- Core studies: Medical Instrumentation, Biomechanics, Physiology, Electronics, Biomaterials & Clinical Engineering.

PROFESSIONAL EXPERIENCE

Technical Manager — *Nov 2024 – July 2025*

Shine Business PLC, Addis Ababa, Ethiopia

- Directed biomedical engineering technical operations, managing client healthcare technology portfolios and service contracts.
- Formulated pre-procurement technical specifications and conducted comprehensive technical and clinical feasibility assessments.
- Supervised clinical equipment installation, acceptance testing, calibration quality assurance, and technical staff training.

Biomedical Officer — *Dec 2023 – Oct 2024*

Central Gondar Zone Health Department, Gondar, Ethiopia

- Led zonal healthcare technology planning and medical equipment management across primary and secondary health facilities.
- Performed comprehensive hospital equipment audits, safety compliance checks, and scheduled preventive maintenance.
- Prepared technical advisories for regional health authorities regarding medical technology procurement and decommissioning.

Biomedical Engineer — *Apr 2022 – Dec 2023*

Amhara Regional Health Bureau (Debre Birhan & Debark Hospitals), Ethiopia

- Delivered hands-on clinical engineering: corrective maintenance, emergency repairs, and calibration of clinical diagnostic systems.
- Executed electrical safety inspections (IEC 60601 protocols), performance validation, and hospital technician training.
- Reduced critical device downtime by streamlining technical service logs and spare-parts inventory tracking.

ACADEMIC & RESEARCH INTERESTS

- Biosignal Processing & Physiological Filtering (ECG, EMG, EEG, Kinematics)
- Ambulatory Gait Analysis & Lower-Back Inertial Sensing (Single-Task & Dual-Task Walking)
- Rehabilitation Engineering & Adaptive Assistive Devices
- Computational Neuroimaging & Non-Invasive Neuromodulation (rTMS)
- Healthcare AI & Explainable Deep Learning for Clinical Decision Support
- Healthcare System Digitalization & Digital Health Telemetry Architectures
- Medical Device Electrical Safety (IEC 60601) & Regulatory Standards Guidance

PROFESSIONAL INITIATIVES & LEADERSHIP

Biomedical Horizon Network (BHN) — Founder & Lead Coordinator

A developing professional and academic network focused on biomedical engineering, healthcare technology, and interdisciplinary collaboration.
Fostering partnerships in medical technology assessment, digital health adoption, and technical capacity building for emerging healthcare systems